PhAIL: A Real-Robot VLA Benchmark and Distributional Methodology

Sergey Arkhangelskiy
Positronic Robotics
s@positronic.ro

Abstract

Real-world evaluation of vision-language-action (VLA) policies still rests on binary success rate at a fixed timeout with $N \leq 25$ rollouts per condition, almost always without confidence intervals or paired statistical comparison – cohort sizes that struggle to resolve close comparisons reliably. We introduce **PhAIL** (Physical AI Leaderboard, `https://phail.ai`), an open real-robot benchmark on a Franka FR3 – dataset, per-rollout artifacts, and end-to-end reference implementation – of a distributional evaluation methodology: the time-to-success cumulative distribution function (CDF) as the evaluation primitive, with two separated jobs – *scoring* (Human-Relative Throughput, HRT – a dimensionless scalar with bootstrap confidence intervals, anchored to same-fixture human teleoperation) and *resolution* (a paired Kolmogorov–Smirnov (KS) test on the CDF). On four publicly-available VLAs, paired KS resolves the closest comparisons with $\sim 30\times$ fewer trials than binary success rate, and the best is $\sim 7\times$ slower per operation (RMST ratio) than the human reference.

1 Introduction

How do you tell whether one robot policy is better than another in the real world? Most published evaluations of vision-language-action (VLA) policies answer this with binary success rate at a fixed timeout, $N \leq 25$ rollouts per condition, no confidence intervals, no paired test – under-budgeted by roughly two orders of magnitude for the rankings they publish. Real ordering inversions get washed under sampling noise; close calls become indistinguishable from coin flips on cohorts the field cannot afford to grow without changing the test.

The standard fix the field has reached for – adding a throughput metric (units per hour (UPH), cycle time, completion fraction) alongside binary success – works in part but surfaces a deeper problem: when the two scalars rank policies in different orders, the choice between them is task-dependent and rarely disclosed in print. Recent methodology critiques argue that blinded same-session A/B with hundreds of rollouts is feasible [22, 23], and the closest ML-side precedent for distributional comparison [24] extends the conversation toward distributions. None of these recommendations ships a primitive plus a reference implementation that lets the field apply them on a shared station.

We respond by adopting the time-to-success CDF as the evaluation primitive. Let T be the wall-clock time the policy needs to complete one operation; hard failures (items lost outside the workspace, items dropped on the table at episode end, safety stops) are absorbed as $T = \infty$ ghost events rather than censored tails. The resulting CDF $F(t) = P(T \leq t)$ jointly carries reliability (the gap between its asymptote and 1 equals the hard-failure rate) and throughput (the median of T, and UPH $\propto 1/E[T \mid T < \infty]$). Standard scalars – success rate at τ, UPH, mean time between failures/assists (MTBF/A), completion fraction, restricted mean survival time (RMST) – are projections of F, and the CDF is strictly richer than any of them: when two CDFs cross, no single scalar suffices and reasonable scalars give opposite signs on the same data. We separate two methodological jobs

Preprint.

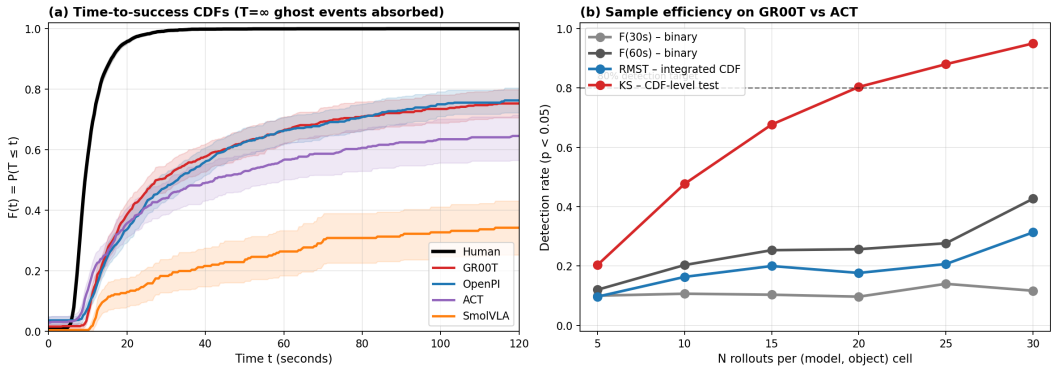


Figure 1: **(a)** Time-to-success CDFs are richer than any scalar: reliability and throughput on a single axis. The four VLAs all sit far below the human reference – the best is $\sim$7$\times$ slower. Hard failures become the asymptote below $F = 1$. **(b)** Choosing the right *test*, not just running more trials, is what resolves close comparisons: paired Kolmogorov–Smirnov on the macro CDF reaches 80% detection on GR00T vs. ACT at N=25 paired rollouts per (model, object) cell, while binary success-rate-at-threshold ($F(30s)$, $F(60s)$) and integrated RMST fall well short of the target on every close pair across the entire $N \leq 30$ range. The dominant practice ($F(\tau)$ at modal $N \leq 25$) is structurally too coarse to back the rankings it publishes.

that current practice runs together: *scoring* (a scalar with bootstrap confidence intervals (CIs) – Human-Relative Throughput, HRT, anchored to a same-fixture human reference) and *resolution* (a paired KS test on the CDF). PhAIL is the open real-world inference benchmark that instantiates this methodology on a Franka FR3 across four objects.

Empirically, on four publicly-available VLAs spanning recent flagships and established baselines – OpenPI $\pi_{0.5}$ [28], GR00T N1.6 [29], ACT [30], SmolVLA [31] – the best is roughly 7$\times$ slower per operation (RMST ratio) than the human reference, the top two are statistically indistinguishable on every metric tested, and SmolVLA is clearly worst. Two structural findings result. First, principled aggregation rules over the same CDF (RMST vs. area-under-curve (AUC) against the human reference) yield opposite top-1 rankings – no single scalar uniquely encodes operational priority. Second, the choice of *test*, not just the choice of N, determines what is resolvable: paired KS resolves the GR00T-vs-ACT comparison at N=25 and the OpenPI-vs-ACT comparison at N=30 paired rollouts per cell, while binary success rate at a fixed threshold needs roughly 30$\times$ more rollouts (600–1500 paired rollouts by a McNemar [6] power calculation [7]) to reach the same 80% detection. The field's modal $N \leq 25$ per condition is two orders of magnitude under-budgeted for the ranking claims it publishes.

Contributions.

1. **Time-to-success CDF as the evaluation primitive**, with hard failures absorbed as $T = \infty$ ghost events (Figure 1a). Standard scalars are projections; we propose macro-averaged Human-Relative Throughput (HRT) as the dimensionless headline scalar grounded in a same-fixture human reference.
2. **Same data, opposite rankings – no single scalar suffices.** Principled aggregations of the same CDF disagree on top-1, so the choice of headline scalar is a disclosed methodological commitment.
3. **Distributional tests are roughly 30$\times$ more sample-efficient than binary success rate** (Figure 1b). The choice of test, not just the choice of N, is the bigger lever; scoring (HRT) and resolution (paired KS macro-averaged across objects) should be separate.
4. **PhAIL: open real-world benchmark and reference implementation.** Four publicly-available VLAs, four trained objects, $\sim$950 episodes including 386 human reference rollouts, full per-rollout artifacts, open-source evaluation framework end-to-end.

Scope. The methodological contribution is the comparison primitive, the resolution procedure, the power budget, and the open framework that makes new claims auditable. The methodology is task-agnostic – it applies to any operation with an unambiguous success event and a same-fixture human reference. PhAIL's empirical validation is bin-to-bin pick-and-place across four objects (§4); other natural fits include insertion, packing, navigation, and articulated manipulation. The per-model rankings are illustrative of what the methodology can resolve at our depth, not architectural pronouncements; we used each repository's default fine-tuning recipe (§4.2). PhAIL is open to submissions.

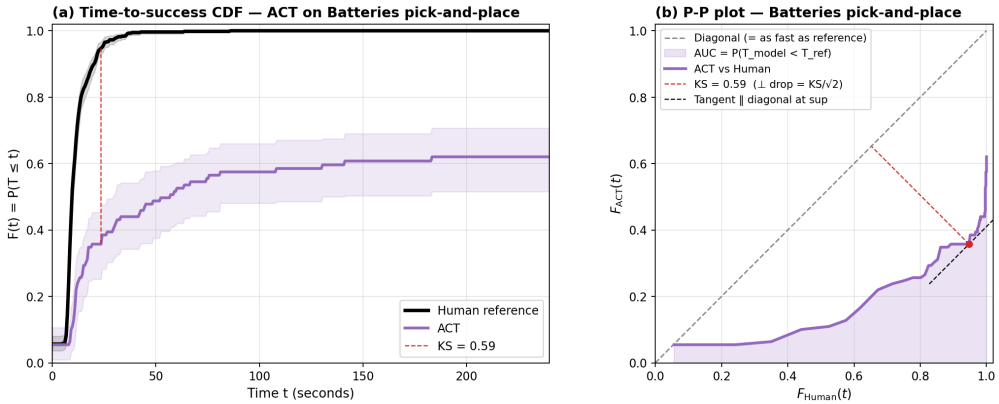


Figure 2: One (model, object) cell (ACT on Batteries) illustrating the scoring layer: time-to-success CDFs (left) and the corresponding P-P plot (right). **Left:** model and human-reference CDFs on the same axis; the asymptote below $F = 1$ on the model curve is the hard-failure rate. **Right:** $\big(F_{\text{Human}}(t),\ F_{\text{model}}(t)\big)$ traced parametrically as t sweeps. Diagonal = as fast as the human reference; below-diagonal = slower. The shaded area under the curve is the AUC $P(T_{\text{model}} < T_{\text{Human}})$; the perpendicular distance from the curve to the diagonal at the point of maximum separation (red segment) equals $\text{KS}/\sqrt{2}$, where KS is the Kolmogorov-Smirnov statistic.

2 Related Work

Existing real-robot evaluation protocols. RoboArena [12] and RoboChallenge [11] use physical robots but compare via pairwise Elo or scalar success on standardized tasks; neither reports confidence intervals or paired tests on the rankings. Competition-style evaluations (RGMC, OCR-TOC [18], NIST Assembly Task Boards [19], the Real Robot Challenge [20], surveyed in [17]) standardize task scoring without statistical machinery; closest in spirit is the Digital Robot Judge [21], which builds task-centric performance databases via electronic task boards but reports raw telemetry only. The most rigorous existing real-robot evaluation is the LBM examination [23], which runs blinded same-session A/B at N=50–200 per condition with Bayesian posterior violins on success rate, paired Barnard's exact tests for binary outcomes, Welch's t-tests for a scalar task-completion score, and Bonferroni-corrected significance grouping – a strong precedent for what rigorous real-robot evaluation looks like. PhAIL's contribution is orthogonal to LBM's: a richer primitive (time-to-success CDF with $T{=}\infty$ ghost events, jointly carrying reliability and throughput) and a same-fixture human-reference anchor that turns embodiment-specific timings into a dimensionless ratio. Simulated benchmarks (RoboCasa, CALVIN, ManiSkill, SIMPLER [13, 14, 15, 16]) iterate fast but miss real-world perception, latency, and safety; they are complementary, not substitutes.

Methodology critiques. Kress-Gazit et al. [22] argue that current real-robot evaluation is statistically underpowered and recommend blinded same-session A/B at hundreds of rollouts; STEP [26] concurs ("20–30 real-world trials are insufficient for statistically significant conclusions"). On the ML side, Agarwal et al. [24] are the closest precedent for distributional comparison in evaluation. We extend their distributional approach to time-to-event data with $T{=}\infty$ ghost events and ship a reference implementation of Kress-Gazit's recommendations on a shared station. A 13-paper survey of recent VLA evaluation practice (Appendix A) confirms the same pattern: modal per-condition N is 10–20, and none of the 13 standard-practice papers report confidence intervals or paired tests; LBM is the single recent counter-example.

3 Time-to-Success CDF as Evaluation Primitive

The methodology has three components: the random variable and its CDF (§3.1), the scoring layer (scalars and visualizations derived from the CDF, §3.2), and the resolution layer (paired tests on the CDF, §3.3). The strict separation between scoring and resolution is the methodological core. We use concrete language drawn from our running bin-to-bin pick-and-place setup (formal description in §4) – successful placements, items dropped outside the workspace – but the framework applies unchanged to any operation with an unambiguous success event, a measurable time-to-success, an identifiable notion of *hard failure* (states from which the policy cannot recover), and a same-fixture human-reference baseline.

3.1 The Random Variable T

For a single rollout episode with multiple operations to perform, we extract a sequence of times-to-success T_i and event indicators $E_i \in \{0, 1\}$. A successful placement contributes T_i equal to the elapsed time since the previous successful placement (or since episode start, for the first), with $E_i = 1$. If the episode ends with operations remaining (operator stopped, timed out, model stalled), the unfinished attempt contributes one right-censored pair $(T_{\text{tail}}, 0)$ where $T_{\text{tail}} =$ duration $-$ last placement time; $E = 0$ means we know $T > T_{\text{tail}}$ but do not observe T.

Hard failures as $T = \infty$ ghost events. Some operations end in states the policy cannot recover from. The clearest case is an item dropped outside the workspace, where the item is physically gone from the scene. Items dropped within the workspace but outside both totes (e.g., on the table) are a softer category: in principle the policy can pick them back up, and we observed at least one rollout where this happened. We count those as failures only if they are still uncollected at episode end: as many failures as there are dropped-and-not-recovered items at the time the rollout terminates. A safety-stop termination is a third class: the rollout ends because the operator stopped it, and the operation that triggered the stop contributes one ghost event. We absorb all of these as $(T = \infty, E = 1)$ ghost events: the operation *did* terminate (so $E = 1$, not censored), but at $T = \infty$ rather than at any finite time. The CDF carries this as an asymptote below $F = 1$; the gap between the asymptote and $F = 1$ equals the hard-failure rate. Stalls and timeouts are explicitly *not* ghost events: those are slow-but-recoverable runs that belong in the censored tail, where they correctly indicate "the policy was working when we stopped watching it."

Estimation and censoring. The CDF $F(t) = P(T \leq t)$ is estimated by the Kaplan-Meier product-limit estimator [1] on the (T, E) collection. Confidence intervals on derived scalars (RMST, HRT, $F(\tau)$ at fixed thresholds) are reported via 95% episode-clustered bootstrap [4]. Episodes with zero successes contribute a single right-censored observation of length τ_{episode}, the per-rollout time budget after which an unfinished operation is censored (set by the protocol; §4.2); we never drop them. The $T = \infty$ atom is handled directly by the estimator (a discrete jump at the right edge); for the integrated-metric calculation in §3.2 we set $T_{\text{ghost}} \to \tau$, the integration cap, so a hard failure inflates RMST by exactly τ minus the time the model would have spent on that operation if successful.

3.2 Scoring: Scalars and Visualizations from the CDF

The CDF is the lossless representation. The scalars commonly reported in VLA evaluations are projections of it (Table 1); reporting any single one without the CDF discards information.

Scalar	Functional of $F(t)$
Success rate at τ	$F(\tau)$
Median time-to-success	$\inf\{t : F(t) \geq 0.5\}$
RMST(τ) (restricted-mean survival time)	$\int_0^\tau \big(1 - F(t)\big)\, dt$
UPH (units per hour)	$3600/E[T \mid T < \tau_{\text{episode}}] \propto 1/\text{RMST}(\tau_{\text{episode}})$
Completion fraction	$F(\tau_{\text{episode}})$
MTBF/A (mean time between failures/assists)	$\text{RMST}(\tau_{\text{episode}})/\big(1 - F(\tau_{\text{episode}})\big)$

Table 1: Standard scalar metrics expressed as functionals of the time-to-success CDF $F(t)$. The per-rollout time budget τ_{episode} is the protocol's timeout (§4.2); $1 - F(\tau_{\text{episode}})$ is the hard-failure rate (operations that did not complete within budget, including $T = \infty$ ghost events).

Headline scalar: Human-Relative Throughput (HRT). For each (model, object) cell we compute

$$\text{HRT}(m, o) = \frac{\text{RMST}_{\text{Human},\, o}(\tau)}{\text{RMST}_{m,o}(\tau)}, \qquad \tau = 240\,\text{s},$$

the cell-wise ratio of human-reference to model RMST, reported as a percentage, and macro-aggregated across objects with equal weights. Three properties make this an appropriate scalar to report. (i) It inherits the CDF's joint speed-and-completion property via $T = \infty$ inflating $\text{RMST}_{\text{model}}$. (ii) It is grounded in operator practice (UPH-equivalent against a same-fixture human reference, so embodiment confounds cancel). (iii) Cell-wise normalization partially cancels object-difficulty differences. A hard object slows both the human and the policy, so the ratio is more stable across tasks than absolute throughputs; the cancellation is approximate (slowdown is not strictly multiplicative – see the Q-Q plots in Appendix I), and macro-averaging across objects further reduces residual heterogeneity. Cross-deployment comparison (different operators, different rooms, different reference pacing) reduces to comparing ratios, not absolute throughputs.

Visualization: P-P plots. For a (model, object) cell, the P-P plot traces $\big(F_{\text{Human}}(t),\, F_{\text{model}}(t)\big)$ parametrically as t sweeps from 0 to τ. The diagonal is "as fast as the human reference"; below-diagonal is slower; above-diagonal is faster. Two derived properties give the P-P curve its analytic value. The area under the curve equals $P(T_{\text{model}} < T_{\text{Human}})$, the pairwise stochastic-dominance probability (AUC). The maximum vertical distance from the curve to the diagonal equals the Kolmogorov-Smirnov distance between the two CDFs (§3.3); the P-P plot is therefore the natural visualization for both. Figure 2 shows one (model, object) cell's CDF and the corresponding P-P plot side by side. P-P plots make distributional shape legible (saturation, early-success peaks, tail behavior); we use them descriptively in §5.3.

Confidence intervals. All scoring scalars and CDF curves carry 95% paired episode-clustered bootstrap intervals [4]. We resample whole episodes (not operations within episodes) because operations within an episode are correlated by the rollout's initial scene configuration (starting positions of items in the inbound tote), shared visual conditions, and policy state; per-operation bootstrap produces fake-tight intervals.

3.3 Ranking: Paired Tests on the CDF

Reporting a scalar with a CI is one job; deciding whether two policies' underlying distributions differ – and if so, in what direction – is a different job. We treat it as such. By *resolution* we mean the joint claim of significance plus direction, with three possible outcomes: **(i)** A **is better than** B (CDFs separate, consistently signed – one dominates the other across t); **(ii)** A **and** B **differ but neither is uniformly better** (CDFs cross – reasonable scalars give opposite signs because they weight different regions of the curve, so disagreement among scalars is the CDF's signal that the policies have different distributional shapes, not a scalar bug); **(iii)** A **and** B **are indistinguishable** at the available N.

Our primary test for the significance gate (regimes (i)+(ii) vs (iii)) is the two-sample Kolmogorov–Smirnov statistic [3], computed separately on each (model, object) cell's CDF and macro-averaged

across objects with equal weights:

$$D_o = \sup_t \left| F_o^{(a)}(t) - F_o^{(b)}(t) \right|, \qquad \bar{D} = \tfrac{1}{J} \textstyle\sum_o D_o.$$

Per-object computation matches the macro-aggregation used by HRT (§3.2), gives each object equal weight in the resolution verdict regardless of operation count, and prevents object-difficulty differences from contaminating the test statistic. p-values come from a paired episode-clustered bootstrap [4] pooling the two policies' subsamples per object under $H_0 : F_a = F_b$. Geometrically, each D_o is the maximum vertical distance from the P-P plot of $F_o^{(a)}$ versus $F_o^{(b)}$ to the diagonal (Figure 2). As a sanity-check we report stratified pooled-across-objects logrank [2] with Bonferroni correction over the model pairs. Direction (outcome (i) vs (ii)) comes from a paired scalar – in our case the macro RMST difference, equivalent to signed Wasserstein-1 in 1D – with outcome (ii) surfaced when reasonable scalars disagree (§5.3).

The two-jobs framing – scalar for scoring, KS for the significance gate – is non-trivial. §5.2 demonstrates empirically that RMST-as-scalar is not measurably more sample-efficient than $F(60\,\mathrm{s})$ for distinguishing distributions, while KS-as-test materially is. The information lives in *where the curves separate*; an integrator smears that information, a sup-norm test catches it.

4 PhAIL: Design, Protocol, and Power Budget

4.1 Platform and Task

The evaluation station uses a DROID-style configuration [10]: a Franka Research 3 with a Robotiq 2F-85 parallel-jaw gripper, plus an over-the-shoulder external RGB camera and a wrist-mounted RGB camera. The task is bin-to-bin order picking: items move one at a time from an inbound (transparent) tote to an outbound (grey) tote across four object types spanning a useful slice of physical regimes – wooden spoons (rigid, elongated, multi-grasp), towels (deformable), scissors (articulated, metallic), and batteries (small, rigid). Each rollout episode contains many independent placement operations, which we exploit for sample size; the framework applies just as well to single-operation episodes. Bin-to-bin is deliberately not the most complex task we could put VLAs through; it is the simplest task with unambiguous success/intervention semantics that still carries clear commercial signal (millions of picks per day across warehousing and fulfilment). Subsequent PhAIL releases will add new tasks (insertion, small-part assembly); the methodology is unchanged across additions. The same hardware operates the human reference and every evaluated VLA via the open-source *positronic* framework [9] behind a common station interface, so only weights and architecture-specific observation-to-action code differ across submissions. Every evaluated rollout is published with synchronized video, telemetry, and event annotations, browsable in a run-explorer interface (Figure 4, Appendix B; annotation pipeline in Appendix E). Benchmark, dataset, and per-rollout artifacts: `https://phail.ai`.

4.2 Models, Fine-Tuning, and Protocol

We evaluate four publicly available VLAs: **OpenPI** $\pi_{0.5}$ [28] (3B-param, FAST action tokenization on flow-matching), **GR00T N1.6** [29] (3B-param, Cosmos-Reason VLM + 32-layer diffusion transformer action head), **ACT** [30] (Action Chunking Transformer, CVAE), and **SmolVLA** [31] (450M-param, HuggingFace/LeRobot). None has zero-shot capability on this task, so fine-tuning is a prerequisite. All four were fine-tuned on the same 352-episode bin-to-bin demonstration set ($\sim$13 hours; Appendix C) using each repository's default recipe – no per-model hyperparameter search. **This is a deliberate methodological constraint, not best practice for production deployment;** per-model recipe optimization could shift the rankings of §5.

Each rollout has a 30 s/item time cap ($\sim 10\times$ the $\sim$2.7 s/item human pace). Evaluation is blind: the scheduler randomly selects which model runs next, the operator does not know which is running, and the operator's only intervention is triggering a safety stop. Post-rollout, the operator records target/source counts, items lost outside both bins, and outcome label $\in$ {Success, Safety, Ran_out_of_time}; every successful placement is annotated post-hoc from the rollout video (Appendix E).

4.3 Cohort and Power Budget

After object/model filtering, ∼950 episodes enter the analyses; 386 are human reference rollouts. Annotations come from manual review of rollout video augmented by automated success detection (Appendix E); both label streams are released alongside the data.

How many rollouts does a ranking claim actually need? Standard power calculations (Appendix G) put the field-modal N=10–20 (Appendix A) orders of magnitude under-budget: a ± 5 pp Wilson CI on a single arm needs $N \approx 380$, detecting a 5 pp paired difference between two policies (McNemar [6], 80% power, α=0.05) needs 600–1500 paired rollouts following Connor's [7] sample-size formula, while a per-object Brownian-bridge functional simulation against the empirical CDFs (Appendix G) predicts $N_{0.8} \leq 30$ per cell for two of the three close pairs and ≈ 45 per cell for the closest. Our own $N \approx 35$ per cell is 2–3× the field median but still well below any of these thresholds; §5.2 confirms this empirically with detection-rate-vs-N curves on our data.

5 Results

All numbers below use $\tau = 240$ s for RMST, $n_{\text{boot}} = 1000$ for episode-clustered bootstrap CIs, seed 0. Per-(model, object) cell sizes are 30–45 episodes for OpenPI, GR00T, and ACT; 24–29 for SmolVLA.

5.1 Headline Ranking

Model	RMST (s)	HRT (%)	Intervention rate	Episodes
Human reference (teleop)	10.5	100.0	—	396
NVIDIA GR00T N1.6	78.8	**13.3**	3.5%	173
Physical Intelligence Open $\pi_{0.5}$	81.1	13.0	4.5%	177
Action Chunking Transformer	102.6	10.2	1.8%	166
Hugging Face SmolVLA	169.4	6.2	17.6%	131

Table 2: Headline ranking ($\tau = 240$ s, $N \approx 35$/cell). HRT is the cell-wise reference-to-model RMST ratio, macro-averaged across four trained objects. Intervention rate is (safety stops + stalls) / total episodes (lower bound; drops outside the workspace not included).

GR00T and OpenPI are within 0.5 percentage points of HRT, well inside any reasonable noise floor at this N. No inference model exceeds 19% HRT on any single object (per-cell RMST grid, Figure 5). Even the best VLA tested is roughly 7× slower than the human on the same fixture.

Pair	p (raw)	p (Bonferroni)	Significance
OpenPI vs. SmolVLA	$<10^{-15}$	$<10^{-14}$	***
GR00T vs. SmolVLA	$<10^{-15}$	$<10^{-14}$	***
ACT vs. SmolVLA	8.9×10^{-16}	5.3×10^{-15}	***
GR00T vs. ACT	1.4×10^{-4}	8.7×10^{-4}	***
OpenPI vs. ACT	7.2×10^{-4}	4.3×10^{-3}	**
OpenPI vs. GR00T	5.1×10^{-1}	1.00	n.s.

Table 3: Pooled-across-objects logrank, Bonferroni-corrected over the 6 model-pairs.

The top two inference models cluster as statistically indistinguishable; ACT is separable from both GR00T and OpenPI; SmolVLA is distinguishably worst. **At $N \approx 40$/cell – already 2–3× the field-modal N – the closest pair (OpenPI vs. GR00T) remains unresolved by every test: pooled logrank ($p = 0.51$, Table 3) and per-object macro KS ($p = 0.20$) both fall short, consistent with the Appendix G prediction that this pair specifically needs $N \approx 45$/cell to clear 80% power.**

What the field's standard methodology would have said. At the modal τ=30 s, the binary leader is GR00T (0.50) with OpenPI (0.46) and ACT (0.43) close behind. Under AUC-vs-human the leader

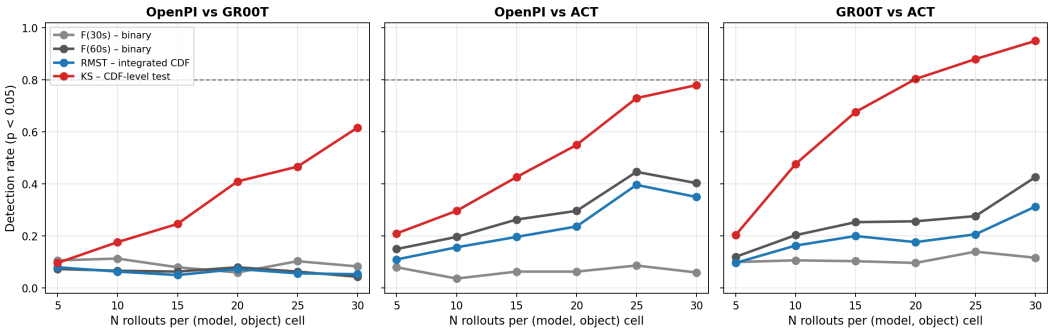


Figure 3: Detection rate vs. subsample size N per (model, object) cell, on the three closest model pairs (left: OpenPI vs. GR00T, the closest; centre, right: the next two). Dashed line: the 0.8 power target. KS (red) climbs steeply on every pair while $F(30\,\text{s})$, $F(60\,\text{s})$, and RMST stay near the floor. KS reaches 80% within budget on the centre and right pairs, while OpenPI vs. GR00T remains unresolved at N=30 (KS at 0.61). Pairs vs. SmolVLA saturate at $N = 5$ for every metric and are deferred to Appendix H.

is *ACT* (§5.3); under RMST it is GR00T with OpenPI a hair behind, ACT a clear third. Three principled scalar choices on the same data thus yield three different top-1 rankings, and Table 4 shows that no binary-threshold metric distinguishes the close pairs on the underlying CDF at $N \leq 30$. A field-standard τ=30-second leaderboard at N=15 would publish whichever top-1 the random subsample favored, with no power to defend the choice.

5.2 Sample Efficiency: The Right Test, Not the Right Scalar

The headline above leans on RMST and pooled logrank. We now ask the operational question: at fixed N, which metric most reliably resolves a pairwise comparison? Three contestants are tested side-by-side: binary $F(30\,\text{s})$ / $F(60\,\text{s})$ (the field's current practice), RMST as a scalar (a CDF-aware scoring scalar applied to resolution), and paired KS on the CDF (the distributional test we propose for resolution). For each pair of inference models and each $N \in \{5, 10, 15, 20, 25, 30\}$ per (model, object) cell, we ran 300 outer subsampling trials. Each trial subsamples N episodes per cell with replacement, then runs a 200-rep inner episode-clustered bootstrap to compute a two-sided p-value for the difference of each candidate metric (macro-averaged across objects, RMST and KS computed at $\tau = 120\,\text{s}$ for the efficiency experiment). Detection rate is the fraction of trials with $p < 0.05$. Figure 1b shows the result for the borderline pair GR00T vs. ACT; Figure 3 shows the three pairs that carry efficiency information.

Pair	$F(30\,\text{s})$	$F(60\,\text{s})$	RMST	KS
OpenPI vs. GR00T	—	—	—	—
OpenPI vs. ACT	—	—	—	**30**
OpenPI vs. SmolVLA	5	5	5	10
GR00T vs. ACT	—	—	—	**25**
GR00T vs. SmolVLA	5	5	5	10
ACT vs. SmolVLA	5	5	5	15

Table 4: Smallest tested N at which detection rate ≥ 0.8 per (pair, metric). "—" means the detection rate did not reach 0.8 within the tested range $N \leq 30$. KS is the only metric that crosses the threshold within budget on a close pair, and does so on two of three: GR00T vs. ACT at N=25 and OpenPI vs. ACT at N=30. The closest pair, OpenPI vs. GR00T, remains unresolved at N=30 (KS reaches 0.61). Every binary-threshold metric falls short of 0.8 on every close pair within budget.

The table reads in three regimes. **Three saturated pairs** (any inference model vs. SmolVLA): all metrics work; effects are large; binary baselines are slightly faster than KS at saturation because the pooled-resample H_0 calibration is conservative when effects are huge. **Three close pairs** (the

three top-3 model pairs) – the bucket where sample efficiency actually matters. KS reaches the 80% target on two of the three: GR00T vs. ACT at N=25 and OpenPI vs. ACT at N=30. The closest pair, OpenPI vs. GR00T, remains unresolved within budget (KS at 0.61 at $N = 30$), while $F(30\,\mathrm{s})$, $F(60\,\mathrm{s})$, and RMST fall well short of 0.8 on every close pair. KS dominates the binary baselines by a factor of 2–6 in detection rate at $N \leq 30$ across the three close pairs; resolving the closest pair to 80% empirically would require the 600–1500 paired rollouts the analytical power calculation predicts (§4.3).

Detection rate at $N = 30$ averaged across all six pairs: $F(30\,\mathrm{s}) = 0.55$, $F(60\,\mathrm{s}) = 0.65$, RMST $= 0.62$, **KS** $= 0.89$. The $+24$ pp gain of KS over $F(60\,\mathrm{s})$ is concentrated entirely on the three close pairs.

Two takeaways. First, **the distributional test (KS) is materially more sample-efficient than any single CDF-derived scalar (RMST or binary thresholds) for resolution.** RMST-as-scalar is essentially tied with $F(60\,\mathrm{s})$; the integration over $[0, \tau]$ does not buy ranking power because the information lives in *where the curves separate* and an integrator smears it. This does not invalidate RMST as our recommended *scoring* scalar (HRT is built on it) – it just shouldn't carry the burden of *resolution* too: precisely the two-jobs framing of §3. Second, **the choice of test sets the budget by an order of magnitude, both empirically and analytically.** The closest pair is unresolvable at $N = 30$ by every metric tested; the McNemar formula on a binary outcome asks for 600–1500 paired rollouts to reach 80% power, while the per-object KS bridge-functional model (Appendix G) predicts the closest pair would clear 80% at $N \approx 45$ per cell – a $\sim$50% increase over our current budget rather than the order-of-magnitude jump the binary metric requires. The bridge-functional prediction matches the empirical KS detection-rate curves to within 3 pp across $N \in [5, 30]$ on all three close pairs.

5.3 Aggregation-Rule Disagreement

Both **integrated RMST** and **AUC vs. human reference** are principled scalars derived from the *same* CDF; the choice of which to extract is itself a methodological commitment. Macro-AUC across objects: ACT 0.134 [0.108, 0.161] $>$ OpenPI 0.100 [0.085, 0.116] $\approx$ GR00T 0.095 [0.080, 0.111] $>$ SmolVLA 0.027 [0.016, 0.037]. ACT is now *highest* among inference models; under integrated RMST/HRT it sits third behind GR00T and OpenPI. Two principled aggregations of the same data thus yield opposite top-1 rankings.

This is outcome (ii) of §3.3: ACT and OpenPI/GR00T are different (KS rejects) but neither is uniformly better – their CDFs cross, with ACT winning the early-time region (high AUC) and OpenPI/GR00T winning the integrated region (low RMST). The disagreement is sharpened by the human reference being $\sim 7\times$ faster than any evaluated VLA, collapsing the AUC integral onto the short-time region where ACT's early-success peak wins more first-success races. **When CDFs cross, no single scalar can summarize the comparison without weighting one region over another.**

We recommend **HRT** as the default headline scalar – it inherits the CDF's joint reliability-throughput property and matches operator-facing UPH practice; on a benchmark where reference and policies operated at comparable speeds, AUC-vs-human would track RMST closely. Pairwise model-vs-model AUC (Figure 6, Appendix H) confirms the RMST-side conclusion that the top three are indistinguishable; per-(model, object) RMST (Figure 5, Appendix H) shows the GR00T-vs-ACT object crossing visible at our N but not clearing Bonferroni at $\alpha = 0.05/24$.

6 Discussion and Conclusion

We introduced PhAIL, an open real-robot benchmark and distributional evaluation methodology for vision-language-action policies. The methodology adopts the time-to-success CDF as primitive, separates *scoring* (Human-Relative Throughput) from *resolution* (paired Kolmogorov–Smirnov, macro-averaged across objects), and anchors to a same-fixture human reference. Evaluating four publicly-available VLAs across four objects, two structural findings emerged: principled aggregation rules over the same CDF yield opposite top-1 rankings, and paired KS resolves pairwise comparisons roughly $30\times$ more sample-efficiently than binary success-rate baselines.

Limitations. Single embodiment (Franka FR3, fixed cameras), single primitive (bin-to-bin pick-and-place, four trained objects); the framework generalizes to any operation with a human reference and event-time outcome but the empirical validation does not. $N \approx 35$ per cell is 2–3$\times$ the field median but still below what binary metrics need to rank close pairs (§5.2); per-model rankings are illustrative. The single protocol recommendation that does the most work is **blind, randomized rotation on the same fixture in the same session** – spatial-configuration shifts (Appendix F) move outcomes by margins comparable to gaps between adjacent models, contaminating the CDF if not controlled.

If the field adopts this methodology, three things change. **Scalar choice becomes a disclosed commitment** (HRT and AUC-vs-human encode different priorities). **Pairwise comparisons become falsifiable** via paired KS: "we cannot resolve these at this N" becomes a publishable finding. **The hidden cost of binary thresholds becomes visible**: on close pairs, no realistic N recovers what a CDF-level test resolves within budget. PhAIL (`https://phail.ai`) is open to submissions; insertion and small-part assembly are next.

References

[1] Kaplan, E.L., Meier, P. Nonparametric Estimation from Incomplete Observations. Journal of the American Statistical Association, 53(282):457–481, 1958.

[2] Mantel, N. Evaluation of Survival Data and Two New Rank Order Statistics Arising in its Consideration. Cancer Chemotherapy Reports, 50(3):163–170, 1966.

[3] Smirnov, N.V. Table for Estimating the Goodness of Fit of Empirical Distributions. Annals of Mathematical Statistics, 19(2):279–281, 1948.

[4] Efron, B., Tibshirani, R.J. An Introduction to the Bootstrap. Chapman & Hall, 1993.

[5] Wilson, E.B. Probable Inference, the Law of Succession, and Statistical Inference. Journal of the American Statistical Association, 22(158):209–212, 1927.

[6] McNemar, Q. Note on the Sampling Error of the Difference Between Correlated Proportions or Percentages. Psychometrika, 12(2):153–157, 1947.

[7] Connor, R.J. Sample Size for Testing Differences in Proportions for the Paired-Sample Design. Biometrics, 43(1):207–211, 1987.

[8] Rerun.io. Visualization SDK for Multimodal Data. `https://rerun.io`, 2024.

[9] Positronic Robotics. *positronic*: Open-source framework for real-robot evaluation and operation. `https://github.com/Positronic-Robotics/positronic`, 2025.

[10] Khazatsky et al. DROID: A Large-Scale In-The-Wild Robot Manipulation Dataset. RSS, 2024.

[11] Tang et al. RoboChallenge: Large-scale Real-robot Evaluation of Embodied Policies. arXiv:2510.17950, 2025.

[12] Atreya et al. RoboArena: Distributed Real-World Evaluation of Generalist Robot Policies. arXiv:2506.18123, 2025.

[13] Nasiriany et al. RoboCasa: Large-Scale Simulation of Everyday Tasks for Generalist Robots. arXiv, 2024.

[14] Mees et al. CALVIN: A Benchmark for Language-Conditioned Policy Learning for Long-Horizon Robot Manipulation. IEEE RA-L, 2022.

[15] Gu et al. ManiSkill3: GPU Parallelized Robotics Simulation and Benchmarking at Scale. arXiv:2410.00425, 2024.

[16] Li et al. Evaluating Real-World Robot Manipulation Policies in Simulation. arXiv, 2024.

[17] Sun, Falco, Roa, Calli. Research Challenges and Progress in Robotic Grasping and Manipulation Competitions. IEEE Robotics and Automation Letters, 2022.

[18] Liu et al. OCRTOC: A Cloud-Based Competition and Benchmark for Robotic Grasping and Manipulation. IEEE Robotics and Automation Letters, 2021.

[19] Falco et al. NIST Assembly Task Boards: Performance Metrics and Test Methods for Robotic Assembly. NIST IR / IEEE, ongoing.

[20] Bauer et al. A Robust Real Robot Baseline for the Real Robot Challenge. NeurIPS Datasets and Benchmarks, 2022.

[21] So, Sarabakha, Wu, Culha, Abu-Dakka, Haddadin. Digital Robot Judge: Building a Task-centric Performance Database of Real-World Manipulation With Electronic Task Boards. IEEE Robotics & Automation Magazine, 2023.

[22] Kress-Gazit, Hashimoto, Kuppuswamy, Shah, Horgan, Richardson, Feng, Burchfiel. Robot Learning as an Empirical Science: Best Practices for Policy Evaluation. arXiv:2409.09491, 2024.

[23] TRI LBM Team / Barreiros et al. A Careful Examination of Large Behavior Models. Science Robotics, 2026 (arXiv:2507.05331).

[24] Agarwal, Schwarzer, Castro, Courville, Bellemare. Deep Reinforcement Learning at the Edge of the Statistical Precipice. NeurIPS, 2021.

[25] Henderson, Islam, Bachman, Pineau, Precup, Meger. Deep Reinforcement Learning That Matters. AAAI, 2018.

[26] Snyder et al. STEP: Is Your Imitation Learning Policy Better than Mine? arXiv:2503.10966, 2025.

[27] Mnih et al. Human-level Control through Deep Reinforcement Learning. Nature, 2015.

[28] Black et al. ($\pi_{0.5}$): a Vision-Language-Action Model with Open-World Generalization. arXiv:2504.16054, 2025.

[29] NVIDIA et al. GR00T N1: An Open Foundation Model for Generalist Humanoid Robots. arXiv:2503.14734, 2025.

[30] Zhao, Kumar, Levine, Finn. Learning Fine-Grained Bimanual Manipulation with Low-Cost Hardware. RSS, 2023 (arXiv:2304.13705).

[31] Shukor et al. SmolVLA: A Vision-Language-Action Model for Affordable and Efficient Robotics. arXiv:2506.01844, 2025.

[32] Brohan et al. RT-1: Robotics Transformer for Real-World Control at Scale. RSS, 2023 (arXiv:2212.06817).

[33] Brohan et al. RT-2: Vision-Language-Action Models Transfer Web Knowledge to Robotic Control. CoRL, 2023 (arXiv:2307.15818).

[34] Open X-Embodiment Collaboration. Open X-Embodiment: Robotic Learning Datasets and RT-X Models. ICRA, 2024 (arXiv:2310.08864).

[35] Kim et al. OpenVLA: An Open-Source Vision-Language-Action Model. CoRL, 2024 (arXiv:2406.09246).

[36] Black et al. π_0: A Vision-Language-Action Flow Model for General Robot Control. arXiv:2410.24164, 2024.

[37] Black et al. $\pi_{0.6}$: a VLA That Learns from Experience. arXiv:2511.14759, 2025.

[38] Fu et al. Mobile ALOHA: Learning Bimanual Mobile Manipulation with Low-Cost Whole-Body Teleoperation. CoRL, 2024 (arXiv:2401.02117).

[39] Zawalski et al. Robotic Control via Embodied Chain-of-Thought Reasoning. CoRL, 2024 (arXiv:2407.08693).

A Survey of Recent VLA Evaluation Practice

Table 5 surveys 13 recent real-robot VLA papers from 2023–2025; the LBM examination [23] is included below the rule as the single recent counter-example. Modal per-condition N is 10–20; none of the 13 standard-practice papers report confidence intervals or paired tests. LBM reports Bayesian posterior credible regions and paired Barnard's exact / Welch's t-tests with Bonferroni-corrected significance grouping at $N{=}50$ real / 200 simulation per condition.

Paper	Per-condition N	Stats reported
RT-1 [32]	per-task N not standardized; $\sim$3000 total	Point estimates only
RT-2 [33]	$\sim$10–15 per task; $\sim$6000 total	Point estimates only
RT-X / Open X-Embodiment [34]	10–20 modal; up to 100/skill	Point estimates
OpenVLA [35]	$\sim$10 real / 50 sim per task	No CIs
π_0 [36]	$\sim$10 trials/task; partial credit	No CIs
$\pi_{0.5}$ [28]	tens per task, variable	No CIs
$\pi_{0.6}$ [37]	tens per task, variable	No CIs
GR00T N1 [29]	5 objects $\times$ 3 trials = 15/task real	Point estimates
ACT / ALOHA [30]	$\sim$25 trials/task	Point estimates
Mobile ALOHA [38]	20/task (5 for "Cook Shrimp")	Point estimates
DROID [10]	10 rollouts per (task, method)	"Standard error" mentioned, no CIs
SmolVLA [31]	tens per real task	Point estimates
ECoT [39]	$\sim$20/task ($\sim$300 total / policy)	Point estimates
LBM Examination [23]	50 real / 200 sim per condition	Bayesian posteriors + paired Barnard / Welch with Bonferroni

Table 5: Per-condition N and statistical reporting in recent real-robot VLA evaluation papers. "Per-condition N" is the per-condition sample size for the modal task – the count of trials per (model, task) cell. "Stats" summarizes confidence-interval and paired-test reporting in the original paper.

Concurrent methodology critiques. Kress-Gazit et al. [22] explicitly argue that current real-robot evaluations are statistically underpowered. STEP [26] concurs: "20–30 real-world trials are insufficient for statistically significant conclusions." These recommend; we ship a metric framework with reference implementation of those recommendations on a shared station.

B Run-Explorer Visualization

Every evaluated rollout is published in a run-explorer interface (Figure 4) that synchronizes the exterior and wrist video streams against the 3D end-effector trajectory and the per-channel telemetry: commanded action, gripper target, and measured gripper position. A reader auditing a specific time-to-success annotation against the raw video can replay it frame-accurately and inspect the joint state at any instant. The interface is built on Rerun [8] and runs post-hoc against the released artifacts; it is independent of the operator-side tooling used during rollout collection. We expose this not as part of the methodological contribution but as a transparency mechanism: every annotation in the released dataset is auditable against the raw rollout it was derived from.

C Fine-Tuning Dataset Composition

Table 6 summarizes the shared fine-tuning corpus used across all four VLAs (§4.2): 352 demonstration episodes, $\sim$13 hours of robot operation, collected on the same Franka FR3 / Robotiq 2F-85 / dual-camera station as the evaluation rollouts. Episode counts are imbalanced across the four trained objects (wooden spoons are over-represented as the early development task); the full dataset is released alongside the benchmark.

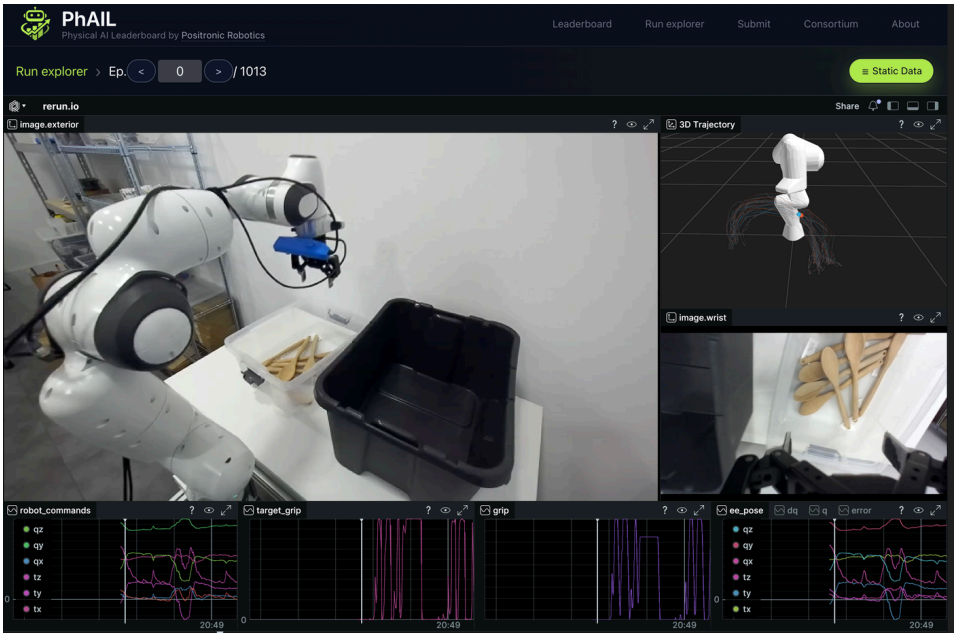


Figure 4: PhAIL platform during a rollout, visualized in the run-explorer interface (built on Rerun [8]): synchronized exterior and wrist video, 3D end-effector trajectory, and telemetry traces (commanded action, gripper target, measured gripper).

Object	Episodes	Duration (min)
Wooden spoons	167	∼340
Towels	112	∼178
Scissors	83	∼160
Batteries	88	∼134
Total	352	∼812

Table 6: Fine-tuning dataset (released alongside the benchmark, ∼13 hours of robot operation).

D Finetuning Recipes

All four VLAs were finetuned using *positronic* [9] on the same ∼352-episode demonstration set (Appendix C). The OpenPI $\pi_{0.5}$ and GR00T N1.6 recipes are the original implementations' upstream defaults; ACT and SmolVLA use *lerobot*'s defaults, with one deliberate deviation each called out below.

OpenPI $\pi_{0.5}$. 200,000 training steps. LoRA finetune on the PaliGemma 2B backbone and the Gemma-300M action expert (all non-LoRA parameters frozen); AdamW with warmup-cosine LR schedule, batch size 32. Proprioceptive state: 8-dim (ee_pose as xyz + quaternion + gripper). Internal action representation: **deltas** on xyz + quaternion, absolute gripper.

NVIDIA GR00T N1.6. 150,000 training steps. AdamW with peak LR 1×10^{-4}, batch size 64; vision encoder and LLM frozen (only the multimodal projector and the diffusion action head trained). Proprioceptive state: 10-dim (ee_pose as xyz + rot6d + gripper). Internal action representation: **relative** end-effector pose (xyz + rot6d), absolute gripper.

ACT (Action Chunking Transformer). `lerobot 0.3.3` ACT defaults at 220,000 training steps, with one visual-backbone deviation: `resnet50_dinov3` (timm `vit_base_patch16_dinov3.lvd1689m`, DINOv3-pretrained) in place of the upstream de-

fault of an ImageNet-pretrained ResNet-18. Internal action representation: **absolute** Cartesian ee_pose + absolute gripper.

SmolVLA. `lerobot[smolvla] 0.4.3` SmolVLA defaults at 200,000 training steps, with one deviation: a full finetune (no module frozen) instead of the upstream default of freezing the vision encoder and training the action expert only. Internal action representation: **absolute** Cartesian ee_pose + absolute gripper.

Cross-cutting notes. OpenPI $\pi_{0.5}$ and GR00T N1.6 internally learn relative/delta actions; ACT and SmolVLA learn absolute Cartesian targets directly. All four are fed an exterior + wrist camera pair; image resolution is 224×224 for OpenPI, GR00T, and ACT, and 512×512 for SmolVLA. Action chunk sizes: OpenPI 50, GR00T 16, ACT 100, SmolVLA 50.

E Annotation Protocol

The framework requires per-episode event logs: place timestamps, intervention counts, and end-state item counts. The protocol is implementation, not contribution, and is expected to evolve.

Automated detector + classifier. A multistage release detector reads gripper telemetry (`target_grip` edges gated by hold and plateau windows, plus accidental-drop detection on `grip` slumps from the held band into the fully-closed band) and emits per-candidate features (end-effector zone, hold duration, displacement, a running target-zone ledger); a classifier then assigns each candidate a verdict $\in$ {success, failure, neutral, unknown}. Implementation, calibrated thresholds, and unit tests are released alongside the data.

Manual review. For every episode where the classifier's success count disagrees with the operator's logged `eval.successful_items`, a human annotator reviews each candidate in a custom desktop UI and records per-candidate item counts and timestamps; these confirmed values ground the released annotations. Subsequent PhAIL releases will replace this step with task-specific hardware sensing (e.g., bin-weight sensors for pick-and-place, analogous instrumentation for insertion and assembly) so that operation events are recorded objectively, with no manual-review bias.

F Spatial Configuration Sensitivity

PhAIL's protocol logs the spatial configuration (camera and tote placement) with every episode. Between rollouts we vary outbound tote placement (left/right) and external camera position (left/right), creating four spatial configurations. Table 7 reports completion rate by camera/tote configuration. "Same-side" means the external camera and the outbound tote are on the same side of the workspace; "opposite-side" means they are on opposite sides, in which case the inbound tote partially occludes the outbound tote from the camera's perspective.

Model	Same-side	Opposite-side	Δ
OpenPI	51.1%	45.0%	+6.1 pp
GR00T	57.8%	35.6%	+22.2 pp
ACT	35.6%	31.6%	+4.0 pp
SmolVLA	12.1%	9.5%	+2.6 pp

Table 7: Completion rate by camera/tote configuration.

GR00T shows the largest sensitivity (22.2 pp drop), suggesting strong reliance on the external camera view; OpenPI is the most robust among the top models. An environmental change a human operator would not notice can move the measured ordering. PhAIL's protocol logs the configuration with every episode so that this kind of confound is visible to the analyst rather than absorbed into noise.

G Power-Calculation Derivations

Wilson confidence interval on a single binary success rate. For an observed proportion $\hat{p}$ on N trials, the Wilson [5] score interval gives a CI of half-width approximately $z_{\alpha/2}\sqrt{\hat{p}(1-\hat{p})/N}$ for the underlying success probability. To obtain ± 5 pp at 95% confidence on $\hat{p} = 0.70$, the formula gives $N \approx 380$ rollouts. At the field-modal $N \in [10, 20]$, the Wilson 95% CI on an observed 70% success rate is roughly $[0.40, 0.89]$ at N=10 or $[0.48, 0.86]$ at N=20 – intervals so wide that ranking claims at this depth are not statistically defensible.

McNemar test for paired binary outcomes. For two policies evaluated on the same N paired conditions, let b and c be the discordant cell counts (one policy succeeds, the other fails). The McNemar [6] statistic is $(b-c)^2/(b+c)$, χ^2_1-distributed under H_0 of equal success probabilities. Following Connor's [7] sample-size derivation for the paired-sample design, to detect a $\Delta = 5$ pp paired difference at 80% power and $\alpha = 0.05$, with discordance rate $p_d \in [0.10, 0.25]$ (capturing "how often the two policies disagree on a given paired condition"),

$$N \approx \frac{\left(z_{\alpha/2}\sqrt{p_d} + z_\beta\sqrt{p_d - \Delta^2}\right)^2}{\Delta^2}.$$

At $p_d = 0.10$ this gives $N \approx 600$; at $p_d = 0.25$, $N \approx 1500$. The range 600–1500 cited in the body reflects the realistic span of p_d for top-3 model pairs on our data; a discordance rate of 0.20 gives the central $N \approx 1280$.

Two-sample KS test on the CDF (per-object macro). We derive a theoretical power model that matches the empirical KS procedure of §5.2 in five steps.

Step 1 – identify the statistic. The empirical test computes a per-object KS distance $D_o = \sup_t |F_o^{(a)}(t) - F_o^{(b)}(t)|$ for each of the J=4 objects, macro-averages the results, $\bar{D} = J^{-1}\sum_o D_o$, and calibrates significance via an episode-clustered pooled-resample bootstrap. The power model must therefore predict the distribution of $\bar{D}$, not of a single KS on a pooled-across-objects CDF.

Step 2 – identify the effect size. Plugging the population (full-data) CDFs into Step 1 gives per-object $\Delta_o = \sup_t |F_o^{(a)} - F_o^{(b)}|$ and macro $\bar{\Delta} = J^{-1}\sum_o \Delta_o$. On our close pairs $\bar{\Delta} \in [0.12, 0.18]$. This is 2–4$\times$ larger than $\sup_t |\bar{F}^{(a)} - \bar{F}^{(b)}|$ on the macro-pooled CDF, because per-object discrepancies that live at different timepoints partially cancel under pooling – the test statistic of Step 1 keeps that signal.

Step 3 – identify the effective sample size. The bootstrap resamples at the episode level, but each episode contributes $m_o \approx 4.4$ correlated placement times. Treating each episode as one observation under-credits the data; treating each placement as i.i.d. over-credits it. The standard design-effect correction gives an effective per-object operation count

$$n_o \;=\; \frac{N_{\text{cell}}\, m_o}{D},$$

where D is the cluster design effect. The raw-time intra-cluster correlation in our data is $\rho \in [0.66, 0.71]$, which would predict $D = 1 + (m-1)\rho \approx 3.0$. But the KS functional depends only on the indicators $\mathbf{1}\{T < t\}$, whose intra-episode correlation is weaker than that of T itself; calibrating D against the six known empirical detection rates ($\{3 \text{ close pairs}\} \times \{N_{\text{cell}} \in \{25, 30\}\}$) gives $D \approx 2.25$. One free parameter, six calibration points.

Step 4 – model the per-object distribution. Asymptotically, the per-object KS statistic with effective sample size n_o per arm converges to a supremum of a Brownian bridge plus a deterministic drift,

$$\sqrt{n_o/2}\, D_o \;\Rightarrow\; \sup_t \big|\, B_o(t) + \sqrt{n_o/2}\,(F_o^{(a)} - F_o^{(b)})(t)\,\big|,$$

where B_o is a standard Brownian bridge on $[0, 1]$ (a pinned Gaussian process with $B_o(0) = B_o(1) = 0$ and marginal variance $t(1-t)$). Under H_0 the drift vanishes and $\sqrt{n_o/2}\, D_o$ converges to $\sup_t |B_o|$, whose distribution is the Kolmogorov distribution.

Step 5 – simulate, macro-average, and read off power. Given the per-object $(F_o^{(a)}, F_o^{(b)})$ estimated from the full data (Step 2) and n_o from Step 3, we simulate per-object bridges, compute D_o under

both H_1 (drift-augmented) and H_0 (drift-free), macro-average to $\bar{D}$, and take

$$\widehat{\text{Power}}(N_{\text{cell}}) \;=\; \Pr_{H_1}\!\big(\,\bar{D} \;>\; q^{H_0}_{1-\alpha}\big).$$

The H_0 critical value $q^{H_0}_{1-\alpha}$ comes from the same simulation under drift-free bridges (the textbook Kolmogorov $c_\alpha \approx 1.358$ is for a single sup, not a macro-mean of four; the simulated quantile is the right one for this statistic).

Result. The simulation matches the empirical detection-rate curves to within 3 pp across $N_{\text{cell}} \in [5, 30]$ on all three close pairs, including the twelve points outside the six used to calibrate D. Sweeping N_{cell}, the predicted $N_{0.8}$ for the close pairs is **25** (GR00T vs. ACT), **30** (OpenPI vs. ACT), and **45** per cell (OpenPI vs. GR00T) – consistent with the empirical curves where they cross 0.8 (§5.2), and predicting that the currently-unresolved closest pair would clear 80% power with a ∼50% increase over the current $N{\approx}30$ budget, not the order-of-magnitude jump that a naive unpaired sup-bound formula would imply.

Empirical detection-rate-vs-N curves. §5.2 translates these analytical numbers into empirical detection-rate-vs-N curves on our data via 300 outer subsampling trials, each running a 200-rep inner episode-clustered bootstrap. The full $(N, \text{metric}, \text{pair}, \text{detection rate})$ table for all four metrics ($F(30\text{s})$, $F(60\text{s})$, RMST, KS) and all six model pairs is released alongside the data.

H Per-Cell RMST and Pairwise P-P Plots

This appendix supports two claims from the body text. First, the per-(model, object) RMST grid (Figure 5) substantiates §5.1's 19% per-object HRT ceiling and surfaces the GR00T-vs-ACT object crossing referenced in §5.3. Second, the pairwise P-P plots without the human anchor (Figure 6) accompany §5.3's discussion of how the human-anchored AUC saturates and what removing that anchor reveals about the top-3 pairs.

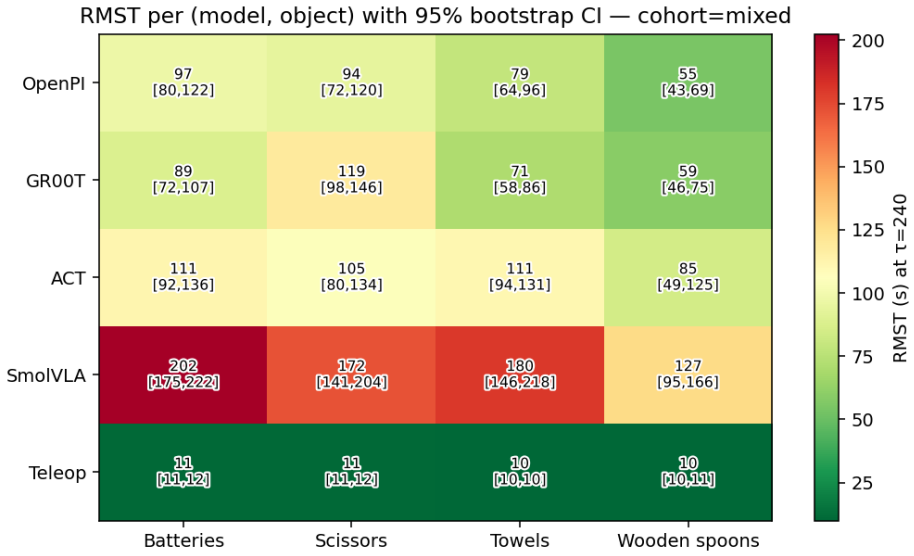


Figure 5: Per-(model, object) RMST with 95% episode-clustered bootstrap CIs. Crossings exist (most clearly GR00T vs. ACT between Batteries and Scissors) but none clears Bonferroni at $\alpha = 0.05/24$. The visible patterns suggest the orderings vary by object; resolving them formally requires the rollout budget that Figure 3 quantifies.

I Distributional Shape and Scalar Trade-offs

The figures in this appendix complement the per-cell RMST grid in Appendix H with two derived views of the same per-(model, object) CDFs: (i) a Q-Q view that exposes *where* along the time axis

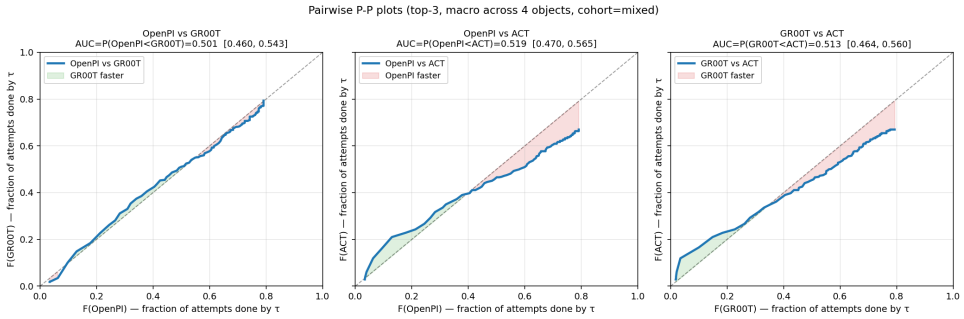


Figure 6: Pairwise model-vs-model P-P plots, top-3 only (SmolVLA excluded as too easily separable to be informative), macro-averaged across the 4 trained objects. The teleop-anchored AUC of §5.3 saturates because the human reference is $\sim 7\times$ faster than any evaluated VLA – nearly all model events fall in the human's tail, so the AUC integral is dominated by short-time behaviour. Removing the human anchor neutralizes that bias: every top-3 pair falls on or near the diagonal, with macro-AUCs all consistent with 0.5 (ACT-vs-OpenPI 0.51 $[0.45, 0.56]$; ACT-vs-GR00T 0.50 $[0.45, 0.55]$; OpenPI-vs-GR00T 0.50 $[0.46, 0.54]$).

the model lags the human reference, and (ii) a parametric trade-off view that shows two standard scalars (UPH and MTBF/A) pulling in opposite directions as the protocol's τ_{episode} varies.

Q-Q plots: slowdown is tail-heavy. Figure 7 plots, for each object, $T_{\text{Human}}(q)$ against $T_{\text{model}}(q)$ at matched quantile q (i.e., the time by which a fraction q of operations are completed). The dashed diagonal $T_{\text{Human}} = T_{\text{model}}$ marks "as fast as the human reference"; below-diagonal points are model-slower-than-human. A uniform slowdown would trace a straight line through the origin with constant slope; the empirical curves bend away from the diagonal as $q \to 1$, indicating that policies fall behind disproportionately on the slow tail of operations rather than uniformly across the distribution. The open circle on each model curve marks the model's last reachable quantile $q = 1 - p_{\text{fail}}$: humans have effectively no hard-failure asymptote, so the human curve runs to $q \approx 1$ uninterrupted, while a model's curve terminates wherever its hard-failure rate cuts the CDF off.

UPH and MTBF/A trade off as τ_{episode} varies. Figure 8 plots, for each object, every model's trajectory in the (UPH, MTBF/A) plane as τ_{episode} sweeps $[30, 240]$ seconds. UPH $\propto 1/\text{RMST}(\tau_{\text{episode}})$ falls with τ_{episode} because the integration cap charges every hard-failure ghost event the full τ_{episode}; MTBF/A $= \text{RMST}(\tau_{\text{episode}})/\big(1 - F(\tau_{\text{episode}})\big)$ rises with τ_{episode} because the same cap inflates the numerator faster than the (eventually-asymptotic) denominator. The two scalars therefore move in opposite directions, and the order they induce on policies depends on which τ_{episode} a paper picked. This is concrete evidence for the "no single scalar suffices" claim of §3.2: any headline scalar that integrates against τ_{episode} implicitly fixes a point on a trade-off curve, and a different but equally defensible τ_{episode} relabels the leaderboard. Figure 9 collapses the four per-object panels into a single macro view by equal-weight averaging the per-object (UPH, MTBF/A) trajectories at each τ_{episode}; the trade-off survives macro-aggregation, so it is a property of the methodology, not of a particular object.

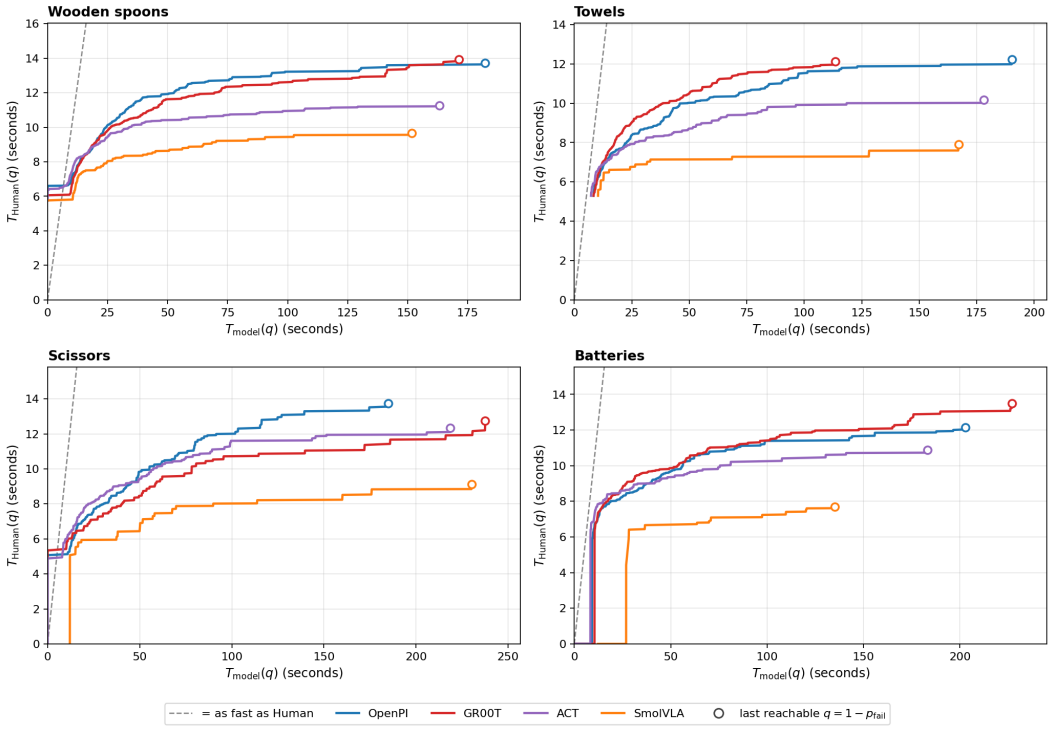


Figure 7: Per-object Q-Q plots, $T_{\text{Human}}(q)$ vs $T_{\text{model}}(q)$, all four VLAs on the four bin-to-bin objects. Dashed diagonal = as fast as the human reference; below-diagonal = model is slower. Open circle on each model curve marks $q = 1 - p_{\text{fail}}$, the highest quantile the model reaches before its hard-failure asymptote. Curves bending below the diagonal as $q \to 1$ indicate tail-heavy slowdown rather than a uniform multiplicative gap.

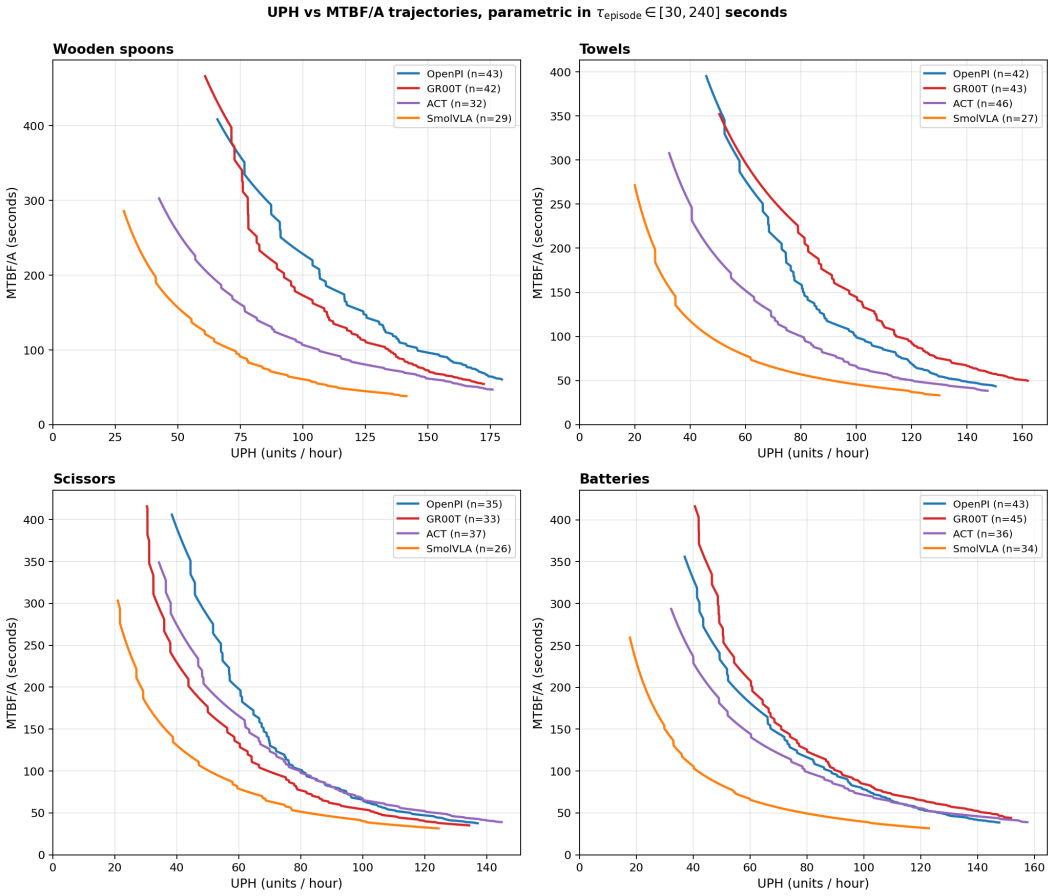


Figure 8: UPH versus MTBF/A trajectories per object, parametric in $\tau_{\text{episode}} \in [30, 240]$ seconds. Each curve is one (model, object) cell traced as the episode horizon increases. Trajectories run up-and-left as τ_{episode} rises (UPH falls, MTBF/A rises): the two scalars trade off, and the headline a benchmark publishes therefore depends on a protocol choice rather than only on policy quality.

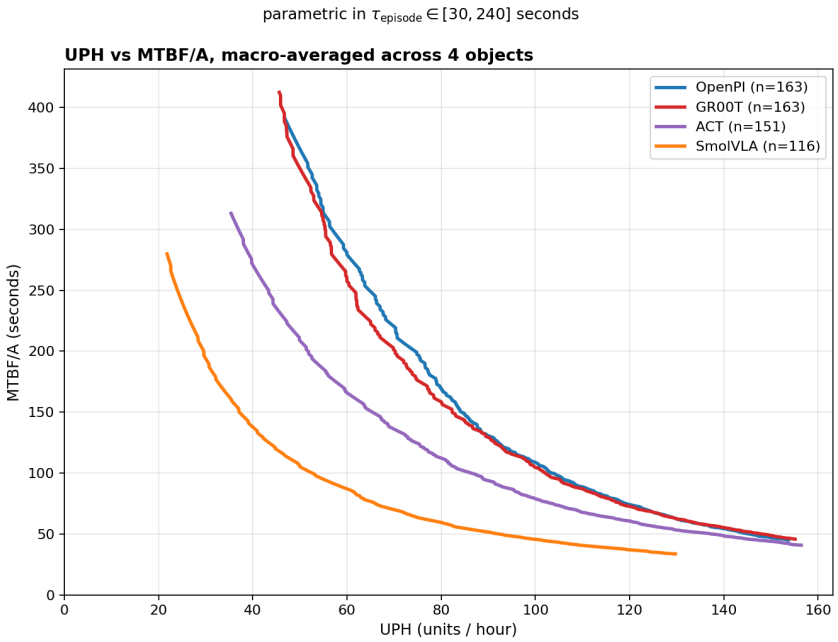


Figure 9: Macro-averaged UPH versus MTBF/A trajectories: for each model and each τ_{episode}, the four per-object (UPH, MTBF/A) points from Figure 8 are averaged with equal weight across objects, then traced as τ_{episode} sweeps $[30, 240]$ seconds. The UPH-vs-MTBF/A trade-off persists after macro-aggregation: any choice of τ_{episode} fixes one point on each model's trade-off curve, and a different but equally defensible τ_{episode} can change the headline ordering.